

Contact

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Top Skills

React.js
Embedded Linux
Amazon Web Services (AWS)

Heather Wisor

Principle Software Engineering Consultant
Bradenton, Florida, United States

Summary

With over 24 years of professional experience, I am a highly skilled and versatile software engineer who specializes in robotic surgery, data visualization, instrument control and embedded software. I have worked with leading companies in the medical device and biotech industries, such as ZAP Surgical, Noah Medical, AbbVie, and Caliper Life Sciences, to design and develop robust and innovative software solutions for complex and challenging problems.

Currently, I am a principle software engineer at Wisor Software, where I have worked on multiple projects from web development, application development and firmware using SQL, C#, .NET, CaliburnMicro, C, C++, Beckhoff I/O modules, EtherCAT, ACS Motion Controller, XAML, WPF, Realtime OSes and more. I am passionate about software design and development, and I enjoy working closely with team members to create successful products that improve the quality of life for patients and clinicians. I am always eager to learn new technologies and skills, and to apply them to solve real-world problems.

Experience

Wisor Software Inc.

Principal Software Engineer at Wisor Software

June 2020 - Present (4 years 11 months)

Lakewood Ranch, Florida, United States

I am a software consultant specializing in both front end and backed end software (Windows, Qnx, Linux and other RTOSes). I provide software development and technical leadership. Have a proven track record of being able to come in and make significant contributions quickly and efficiently. Have designed and developed products for medical device companies such as Zap Surgical, Noah Medical, AbbVie, Perkin Elmer, Nellcor and Puritan Bennett

Noah Medical

Software Engineering Consultant

August 2020 - Present (4 years 9 months)

San Carlos, California, United States

Worked with team to design and implemented AWS cloud architecture, including Lambda functions in C#, StepFunctions, SNS, SQS, React.js, CloudFormation, and SAM to orchestrate log processing and data visualization solutions.

Developed a web-based dashboard with interactive visualizations, leveraging Python, Pandas, Streamlit, Plotly, and Bokeh to analyze and present internal instrument data stored in a SQL database.

Contributed to the development of the Galaxy Surgical Robotic System, working on MVVM UI design, instrumentation software, system architecture, and embedded software. Utilized C#, .NET, WPF, and CaliburnMicro for UI development and C, C++ for low-level system control. Integrated Beckhoff I/O modules, EtherCAT, QNX, FRI, and KUKA Robot for precise robotic control and automation.

USA Firmware

Senior Software Engineering Consultant

February 2024 - February 2025 (1 year 1 month)

Ohio, United States

Developed device drivers and HAL layer on embedded Linux boards with Armbian OS. Developed linux audio driver, fan control, temperature controls, integration of GPIOs, CAP touch buttons, power control and others. Debugging firmware / hw issues using logic analyzer and multi meter.

ZAP Surgical Systems, Inc.

Software Engineering Consultant

October 2020 - January 2024 (3 years 4 months)

San Carlos, California, United States

Redesigned motion control software for ZAP-X Surgical Robotic System containing 7 axes surgical robot, LINAC, X-Ray Imaging, Safety IO modules. ACS Motion Controller, Beckhoff, EtherCAT, C#, C++, PLC

Work with Treatment Delivery Software Engineering team supporting new features for ZAP-X. #C#, #C++, #XAML, #WPF

Nodexus Inc.

Software Engineering Consultant

October 2022 - May 2023 (8 months)

Newark, California, United States

UI Software Engineer focusing on the front-end development of embedded linux QML/QT application. QML, QT, C++, Debian Linux

Paramit Corporation

Lead Software Engineer , Paramit Corporation Product Development Group

June 2019 - June 2020 (1 year 1 month)

Morgan Hill

Lead team in development of GUI and Instrument Control Software application (C++/QT), driving the software architecture, implementation, planning, documentation, and design verification of instrument software for a DNA Printer (see DNAScript / SYNTAX). Participated in both management and as an individual contributor to the software development for this project. This product incorporates complex electro-mechanical components, such as: plate robot, gantry robot, plate reader, valves, many pumps driven by stepper motors, US sensors, multiple rotary valves; and custom hardware / firmware using ST Micro's. Single Board Computer running Ubuntu 18.04 Linux.

Participated in management and development of a prototype instrument consisting of complex electro-mechanical components used to drive fluid through many valves using many pumps in order to deliver uL control to any 96 wells in a plate. Hardware architecture consisted of off the shelf components such as Trinamic motor controllers, Berker valves, etc. This instrument was controlled from a Windows PC using Python .

Allergan

Associate Director of Software Engineering

April 2016 - June 2019 (3 years 3 months)

Pleasanton CA

Associate Director, Software Engineering

Manage Windows software team as well as technical lead. Develop and implementation of reusable \ configurable software architecture and ecosystem for next generation of CoolSculpting product lines. Ecosystem includes complex secure authentication scheme, object-oriented state-driven application code, custom Azure Agent for communicating over a cellular modem to Azure Platform, JSON based logging, custom secure installer, and WPF MVVM UI. Embedded software runs on Windows 10 IOT using a TPM2.0 and Windows security (BitLocker & Device Guard); tools were mainly

PC based and run on Windows 10. All software was implemented using C# and .NET framework.

Author, contributor and reviewer of many software documents (Software Architecture, Software Strategy Plan, Software Design, Coding Standards, Software Development Process, Software Estimate Guidelines, Software Requirements, Verification Test Protocols and Coding Inspections)

ZELTIQ

Sr. Staff Software Engineer

September 2013 - March 2016 (2 years 7 months)

Pleasanton CA

Sr. Staff Software Engineer (9/2013 - 3/2016) (Zeltiq was acquired by Allergan)

Technical lead for all Windows based software used in CoolSculpting product as well as all service, manufacturing and R&D tools. Embedded software ran on WindowsCE 5.0 using object-oriented event driven state machine based architecture, tools ran on Windows 7/10 and utilized a plugin-based architecture using WPF, WCF and Caliburn Micro; C# language. Log files were generated using a custom protocol as well as in JSON format for the PC based tools. Communication protocols included: RS232, Axeda Agent to Axeda Cloud Platform using a REST API. Axeda Cloud Platform supports remote software deployments and log file transfer over a cellular modem. Extensive knowledge of Agent / Cloud Platform communication and remote software deployment capabilities.

Caliper Life Sciences

Staff Software Engineer

July 2010 - August 2013 (3 years 2 months)

Lead architect and embedded software developer for automation controller of new MicroFluidic DNA separation instrument. Designed and implemented embedded software for automation control board running a F28335 DSC, interfacing to Spartan FPGA, and peripherals: USB, SPI bus, configuring GPIOs, stepper motor control, laser, pressure pumps, optics and high voltage control. Wrote optical auto focus routine. Write software requirements, design documentation, and test plans. Code is written in C and C++.

Converted legacy COM based service responsible for communication between application and firmware over USB to a managed library using CLI wrappers.

Implemented C# server devices to mirror embedded software device control. Server code uses a WCF interface to communicate with windows client devices for MicroFluidics DNA separation application.

Developed control and communication software for IVIS CT imaging instrument running F2812 DSP. Worked closely with Electrical engineer to bring up new motor control board with FPGA which served as an interface to stepper motors, DACs, temperature control, and additional IOs. Embedded code implemented in C/C++. Communication software involved a C++ COM server and C# client device software to provide interface for Living Image application software to control devices.

Developed test application written in C# for use in bringing up imaging board, which is now used by Manufacturing to test IVIS platform before shipping to customer.

New features and sustaining of embedded software for LabChip product lines.

Terumo

Sr. Software Engineer

March 2009 - May 2010 (1 year 3 months)

Solely responsible for product UI platform for blood separation product which included architecting and implementing UI software for Atmel ARM AT91R4008 microprocessor, serial and touch screen drivers, interface and implementation of peripheral board software used for printing and for data export to compact flash and USB flash drive. UI platform code written in C, peripheral BitsyXb board used for printing and data capture written in C++ .Net and runs on Windows CE OS 5.0.

Work closely with process and marketing teams to develop software requirements, design documents and UI screens, interaction diagrams, process flow charts, and operating manual.

Support electrical engineers in replacing RAM and FLASH parts due to part obsolescence. Bring up of these chips with existing board architecture, re-writing of memory map and EBI used by microprocessor. Use Oscilloscope and ICE to debug and test. C and Assembly code.

Implement and sustain PC support software application used by manufacturing and engineering team to monitor and run various sensors, pumps, setting

initial product parameters and to also program ELO touch screen chip with initial settings and to then calibrate touch screen and store calibration data for use by UI during normal run mode. PC software written in Visual Basic, communicate to device via RS232 to device.

Company was shut down in 2010.

OptiScan BioMedical Corp.

Sr. Software Engineer

November 2005 - March 2009 (3 years 5 months)

Tyco Healthcare

Sr. Software Engineer

February 2000 - November 2005 (5 years 10 months)

Education

Eastern Washington University

Bachelor of Science - BS, Electrical and Electronics

Engineering · (1995 - 1999)